Tanay Pratap Singh

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Education

Master's of Science in Applied Data Science	**Syracuse University, Syracuse, NY**	Jan. 2025 – Dec. 2026
B.Tech in Computer Science and Engineering	**JSSATE Noida, UP, India**	Dec. 2020 – May 2024

Experience

Research Assistant — Feb. 2026 - Present
Syracuse University — *Syracuse, NY*

- Parsed 276,000+ Zooniverse volunteer classifications for the project Gravity Spy 2.0 (NSF-funded) using Python and pandas, reconciling 5 undocumented metadata schemas across 25,104 LIGO glitch subjects and 8,293 auxiliary channels.
- Reverse-engineered the Zooniverse Panoptes API to build a batch pipeline retrieving 49,000 spectrograms; processed 1200x1200 mosaics into paired 224x224 channel heatmaps with OpenCV.
- Trained a two-input CNN with a shared MobileNetV2 backbone in TensorFlow/Keras to predict volunteer similarity labels from paired strain and auxiliary-channel spectrograms.
- Built a glitch-type by auxiliary-channel similarity matrix from model outputs to identify detector subsystems (PEM, SUS, LSC, ISI, ASC, CAL) causally linked to specific glitch types.

LSS Consultant — Jan. 2026 - Present
JMA Wireless — *Liverpool, NY*

- Designed an executive Power BI dashboard comparing JMA's two Jumper assembly lines (Jumper 1 manual, Jumper 2 fully automated), consolidating 4 operational KPIs into a single decision support view for the SVP of Global Operations.
- Built a Python ETL pipeline to flatten nested JSON PLC records across 500,800+ rows, cutting Power BI memory load from 500+ GB to under 5 MB and enabling one click refresh for non technical users.
- Authored 10+ DMAIC artifacts (Charter, SIPOC, CTQC, FMEA, RASIC, Control Plan) and applied Minitab to baseline process performance.
- Partnered with a four person cross functional team and presented findings to JMA executive sponsors, translating analysis into business recommendations tied to the financial plan.

Cybersecurity Teaching Assistant — Summer 2025
Syracuse University — *Syracuse, NY*

- Developed and delivered 15+ hands-on cybersecurity labs covering on cryptography, network security, penetration testing, and digital forensics using Wireshark, Kali Linux, Nmap, OpenSSL, and Metasploit, which increased student confidence in applying security tools.
- Built and maintained multi-VM lab environments (Kali Linux, Windows 10/Server, CentOS, Metasploitable) to enable realistic attack simulations and vulnerability assessments, allowing students to practice exploitation techniques in a safe setting.
- Graded and reviewed security labs and incident analysis reports, reinforcing packet analysis, OSINT reconnaissance, ethical hacking and hashing (SHA-256, SHA-512) concepts, which helped students achieve clearer understanding of compliance requirements.

Data and Product Analyst Intern — Jul. 2023 – Aug. 2023
ProProfs — *Noida, UP, India*

- Performed customer and product analytics on 2,000+ client datasets using Python, SQL, and Excel to uncover behavior patterns and conversion drivers, enabling the product team to prioritize two high-impact features that lifted trial-to-paid conversion.
- Built Tableau dashboards visualizing feature usage and conversion trends across SaaS products serving 5,000+ users, surfacing under-used features that reduced churn by 5 percent and helped lift product ratings from 4.2 to 4.6.

Technical Skills

Languages and Frameworks: Python, SQL, R, JavaScript, HTML/CSS

ML and AI: Scikit-learn, TensorFlow/Keras, XGBoost, CatBoost, OpenCV, SHAP; Random Forest, Gradient Boosting, LSTM, CNN, NLP, Computer Vision, Feature Engineering, Bayesian Optimization, Model Deployment

Data and Analytics: Pandas, NumPy, Spark, Tableau, Power BI, Excel, Streamlit; EDA, A/B Testing, Time Series Forecasting, Hypothesis Testing, Statistical Analysis

Cloud and Data Engineering: Azure, MySQL, ETL Pipelines, JSON Data Engineering, Git/GitHub, Jupyter

Statistics, Modeling and Business: Linear/Logistic Regression, Multiple Regression, ANOVA, Hypothesis Testing, Time Series Forecasting, Multinomial/Ordered Logit, Conjoint Analysis, K-Means Clustering, Lean Six Sigma (DMAIC), SIPOC, Process Mapping, A/B Testing, Technical Documentation, Stakeholder Communication